Ahmed Khalid Humoudi

Robotics and Automation Engineer

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PROFILE

Dedicated mechatronics engineer with hands-on experience in robotics, control, automation, AI integration, and system development. Passionate and seeking opportunities to apply technical expertise to real-world applications.
Portfolio: portfolio-omega-ten-g6ozj4wxdx.vercel.app - contains detailed project information, visualization, and implementation.

EDUCATION

German International University (GIU), *Bachelor Degree, 2026* 09/2022 – 07/2026 | Cairo, Egypt

Major: Robotics and Automation
GPA: 0.87 (GIU System), equivalent to **3.85** on the United States 4.0 GPA scale.
Ranked among the top 3 students in the Robotics Major.

Global International School (Al-Kon), *High School Degree, 2022* 09/2019 – 06/2022 | Jeddah, Saudi Arabia
IGCSE Score: 120%

PROFESSIONAL EXPERIENCE

Automation Engineering Intern, *Mactech Automation Solutions* 07/2026 – 08/2026 | Jeddah, Saudi Arabia
Developed and tested PLC programs using Siemens TIA Portal on real S7-1500 PLC hardware and ET 200SP I/O systems, alongside PLCSIM for simulation and validation. Worked with LAD programming, Function Blocks, digital and analog I/O, PID control, HMI development, and PROFINET communication.

Mechatronics Engineer Intern, *Hiryo* 02/2026 – 07/2026 | Cairo, Egypt

Designed and developed a Bluetooth-based alarm and tracking device with real-time alert and location features, integrating embedded hardware, firmware, and mobile connectivity to deliver a reliable end-to-end system.

Physics Lab Assistant, *German International University* 02/2024 – 06/2024 | Cairo, Egypt

Conducted hands-on laboratory supervision and technical guidance, assisting in experimental procedures and the practical application of theoretical concepts.

PROJECTS

Bachelor Thesis
Reinforcement Learning-Based Path Planning for Warehouse Robots in Dynamic Environments:
Developed an RL-based navigation system for autonomous warehouse robots operating in dynamic environments with obstacle avoidance and adaptive path planning.

Multi-Robot Task Formation: Centralized ROS2 system, obstacle avoidance, path planning, formation control.

PLC-Based Stepper Motor Motion Control: controlling stepper motor speed, direction, position control.

UR5E Pick and Place: UR5e robot integrated with ROS 2, MoveIt 2, and RViz for trajectory planning.

Oven Control System: ATmega32, temperature monitoring, fan control, and safety lock system.

Ball and Beam Control System: PID-based ball stabilization on a beam using Arduino Uno.

4-DOF Robotic Arm: Arduino-based 4-DOF arm with inverse kinematics and trajectory planning.

TECHNICAL SKILLS

Industrial Automation: Siemens TIA Portal, Rockwell Studio 5000, PLCSIM, WinCC HMI, PROFINET, PID
Robotics: ROS 2, MoveIt 2, RViz 2, Gazebo, MATLAB, Simulink
Programming: C, C++, Python, JavaScript
Embedded Systems: Arduino, Raspberry Pi, ATmega32, ESP32
Artificial Intelligence: Machine Learning, Deep Learning, Reinforcement Learning
Microsoft Office: Word, Excel, PowerPoint
Languages: Arabic (Native), English (C1), German (A2)

COURSES

From Wire to PLC: A Bootcamp in Industrial Automation, *Udemy — Online*

PLC Programming, *HA Consulting Group — In Person*

Modern Robotics: Foundations of Robot Motion, *Northwestern University, Coursera — Online*

Introduction to Artificial Intelligence, *IBM, Coursera — Online*

MATLAB, Simulink, Reinforcement Learning Onramps, *MATLAB Self-Paced Courses — Online*